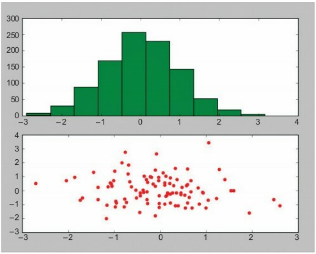

Figure 7: Histogram and scatter plots in Matplotlib

```
pt.show( )
```

The line of code:

```
a = np.random.randn(1000)
```

…generates 1000 random numbers and stores them in the variable 'a'. Similarly, the next two lines of code generate 100 random numbers each, and store them in the variables 'b' and 'c'. This program also uses two sub-plots which we have seen earlier. The line of code:

```
pt.hist(a,color='green')
```

…plots a green histogram using the numbers in variable 'a'. The line of code:

```
pt.scatter(b,c,marker='o',color='red')
```

…plots a scatter plot with the numbers stored in variables 'b' and 'c'. On executing the program, *plot6.py* will give the image shown in Figure 7 as the output.

Though a number of important topics like 3D plotting using Matplotlib, Matplotlib tool kits, etc, have been left out, I am sure this introduction will motivate researchers and professionals into accepting Matplotlib as a powerful tool for scientific visualisation. Earlier, when we discussed the different output formats of Matplotlib, we came across a format called *pgf*. This is the format by which Matplotlib provides PGF/TikZ code to LaTeX. So, in the next article in this series on scientific graphics visualisation, we will discuss PGF/TikZ, yet another powerful graphics visualisation tool. END

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